Analysis and Design of Algorithms

NUS National University of Singapore

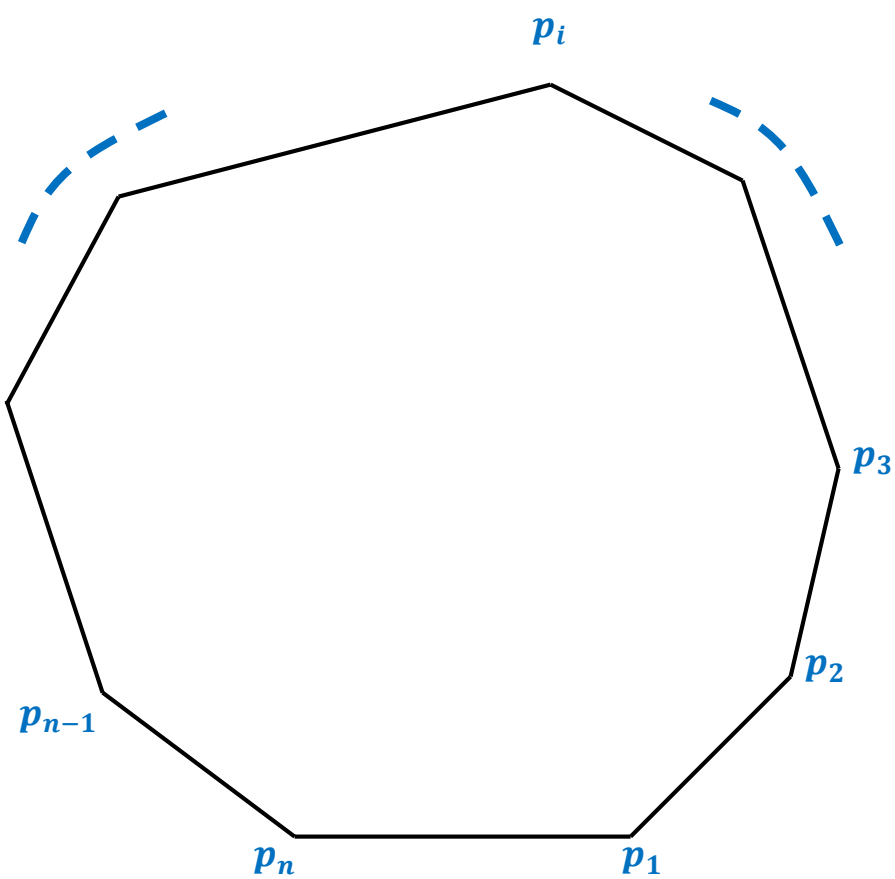
Algorithms

CS3230

Tutorial

Week 9

A Convex Polygon



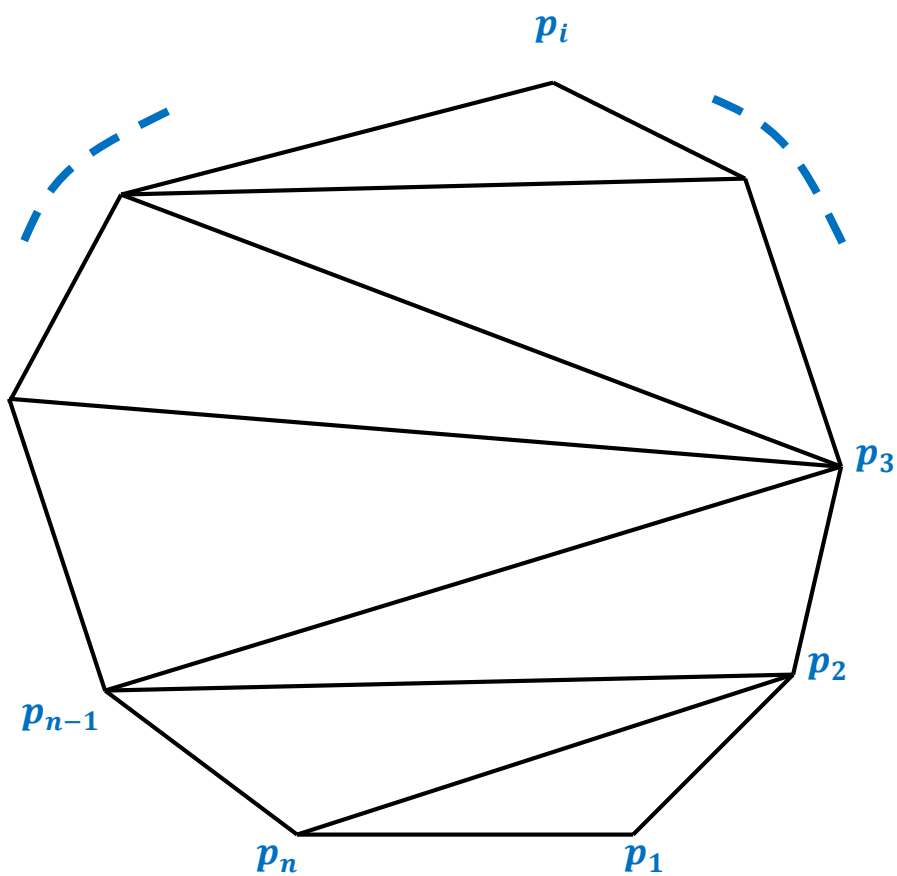
Representation:

< p_1 ,…, p_n > Stored in an array.

< p_i ,…, p_j > :

Polygon consisting of points p_i ,…, p_j

Triangulation of A Convex Polygon

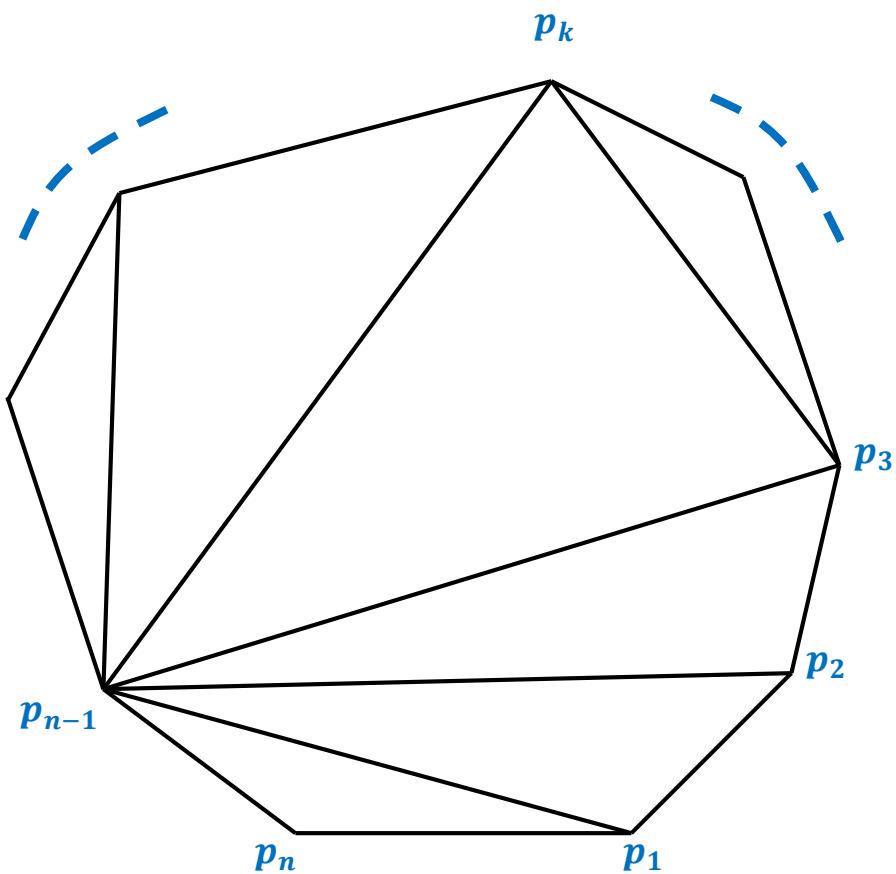


$\omega(i,j,k)$: weight of triangle formed by p_i, p_j, p_k.

Assumption: It takes **O**(1) time to compute $\omega(i,j,k)$

Cost of a triangulation : sum of the weight of $n-2$ triangles formed.

Triangulation of A Convex Polygon



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Question

Problem: Given a convex polygon represented by < $\boldsymbol{p_1}$,…, $\boldsymbol{p_n}$ >, the objective is to find a triangulation with minimum cost.

Design a dynamic programming algorithm for the above problem, that runs in $O(n^3)$ time.